Distinct Gathering and the Virtue of Self-Consistency*

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Abstract

We resolve one of the longest-standing open problems in the field of autonomous mobile robots, disproving the established conjecture, and refute the natural belief that self-consistency cannot help the robots coordinate.

Distinct Gathering is the fundamental cooperation task of letting robots, initially scattered across the plane, meet in a single point. We assume the natural *Round-Robin* scheduler, which activates the robots in turns. An activated robot perceives the locations of all other robots in its local coordinate system and then moves to a destination based on this information.

For $n = 2$ robots, the task is trivial. What happens for $n \geq 3$ has remained an open problem for decades, the established conjecture claiming it to be impossible. Our first main result is that the task is indeed infeasible for $n = 3$ but feasible again for any $n \geq 4$. We provide a robust, uniform algorithm that works even when robots may be adversarially stopped during their movements.

Our second main result challenges another belief that has shaped the field of mobile robots since the beginning. It is natural to see self-consistency as a purely robot-internal property that cannot help robots solve coordination tasks. However, we refute this assumption by showing that self-consistency with respect to specific properties is sufficient to render the problem solvable. Specifically, we prove that Distinct Gathering under Round Robin for $n = 3$ robots, which is unsolvable in the absence of any consistency guarantees, becomes feasible with either of the minimal assumptions of only a self-consistent unit length or only a self-consistent compass. We provide explicit robust algorithms for both cases.

This settles all open questions about Distinct Gathering under Round Robin and corrects the consequential misconception that self-consistency is useless for collaboration.

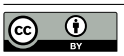
CCS Concepts

• **Computer systems organization → Robotic autonomy**.

Keywords

Autonomous Mobile Robots, Distinct Gathering, Round Robin, Disorientation, Self-Consistency

*A full version of this paper is available at arxiv.org [17].
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1 Introduction

The computational power of autonomously acting, simple, mobile robots has been the object of intense research in the field of distributed computing. Ever since Suzuki and Yamashita's seminal work [27], a large amount of work has been dedicated to the research of theoretical models of such autonomous mobile robots [1, 3, 5, 10, 18, 21, 26]. The standard textbook by Flocchini et al. [13] provides an excellent overview of the basic models and literature. In the standard model, each robot is represented as a point in the two-dimensional plane with limited capabilities. In particular, the robots are assumed to be *oblivious* (have no persistent memory), *anonymous* (have no IDs), and *uniform* (run identical algorithms) [13].

The robots operate in *Look-Compute-Move* (*LCM*) cycles. A robot performs such a cycle whenever it is activated by the so-called scheduler. For the standard model of a *semi-synchronous* scheduler (often denoted Ssync), these activations happen in *rounds*. At the start of each round, the scheduler activates a nonempty set of robots, possibly all. The activated robots then simultaneously perform an *LCM* cycle. Each cycle is composed of three operations: in the *Look* operation, a robot obtains a snapshot of the plane showing the positions of the other robots relative to its own position; in the *Compute* operation, it executes its algorithm (which is identical for all robots) using the snapshot as input; and then it moves toward the computed destination in the *Move* operation. The next round begins only once all robots have stopped their movements. (The model variation without this restriction is called asynchronous and denoted Async.) The scheduler is adversarial but must be *fair*, i.e., activate each robot infinitely often. The repeating cycles then allow the robots to collectively perform various tasks and solve a range of coordination problems.

The scheduler can be restricted in various ways. The fully synchronous scheduler (Fsync) activates all robots simultaneously in every round [13], representing the simplest form of scheduling. The second most basic scheduler is arguably the *Round-Robin* scheduler (often denoted RoundRobin), which activates all robots in turns, keeping the same order forever. RoundRobin is a centralized scheduler, that is, in each round only one robot is activated.

The task of gathering all robots at a single, arbitrary point is fundamental and has been extensively studied. By now, a large collection of possibility and impossibility results for different assumptions is known, many in the base model [4–6, 19, 20, 22, 24, 27] (with small variations, e.g., regarding multiplicity detection) and

also many significantly modified models considering limited visibility [2, 14, 26] or fault tolerance [1, 3, 9] or lights [7, 15, 23], which represent communication and memory capabilities. A recent survey [16] on Gathering reviews and extends the known results in the model relevant here and provides a useful introduction. Gathering is trivial for Fsync: each robot moves to the center of gravity, where they meet after a single activation. This strategy fails for RoundRobin: the robots will converge but never meet. Instead, the easy solution is to target a point of maximum multiplicity [27, Thm. 3.4]. Without multiplicity detection, however, gathering multiple robots becomes impossible already [25, Sec. 3]. We outline here the general proof concept: Consider three robots in two locations. The single robot must move to the opposite location when activated to gather. If an accompanied robot is activated first, however, it has the same view as the single robot (i.e., for an adversarially rotated map they see robot locations at exactly the same coordinates) and therefore also moves to the opposite observed location. This can lead to the robots swapping places forever. Note that this proof relies on an initial configuration where two robots are co-located. This argument fails when we assume that the robots all start from distinct locations. This natural problem variant is called *Distinct* Gathering. (This variant was in fact the default for a long time. Accordingly, the term Gathering without further qualifier used to mean Distinct Gathering. The variant without the distinctness assumption was and sometimes still is referred to as *Self-Stabilizing* Gathering instead.)

Motivation and Contribution. Determining the feasibility of Distinct Gathering under the Round-Robin scheduler has been a long-standing open question in the field. It was raised two decades ago at the International Symposium on Distributed Computing [8] and has been tackled repeatedly. Impossibility could be proven at least for two very similar models: Gathering under the Round-Robin scheduler and Distinct Gathering under the so-called 2-bounded centralized scheduler (see [8]). It was hypothesized that the problem remains impossible for the intersection of the two models, i.e., Distinct Gathering under RoundRobin (which is essentially the 1-bounded centralized scheduler). The most recent attempt at an upper bound is an algorithm that solves the problem at least with the additional assumptions that the robots have some internal memory, rigid movement, and share a unit distance (i.e., all perceive the same consistent unit length) [28, Sect. 4, Alg. 10]. Nevertheless, the general consensus has remained that Distinct Gathering is most likely impossible under RoundRobin without additional assumptions. A conjecture by Défago et al. formalized this hypothesis [9, Conj. 2]. The authors corroborated their belief by proving the impossibility under the plausible assumption that any algorithm solving this problem is a so-called *rapid-gathering* algorithm [9, Conj. 1]. We disprove both conjectures, demonstrating that Distinct Gathering under RoundRobin is impossible for three robots but becomes feasible again for four or more robots.

Moreover, we provide a particularly robust algorithm that only uses straight movements and still works when faced with *nonrigid movement*. The latter means that any robot may be stopped at arbitrary times after moving some unknown minimal movement distance (e.g., due to overheating or a need to recharge depleted batteries). This causes it to lose all of its memory and leaves it to wait for its next activation by the scheduler.

For our second main result, we reconsider the case of three robots. The problem is unsolvable in the default model without any consistency guarantees, as stated above. We show that it becomes feasible again by assuming only a self-consistent unit or a self-consistent sense of direction. This result disproves the natural belief that self-consistency, which is purely robot-internal, cannot help the robots cooperate. This underscores the importance of precision in model descriptions when it comes to assumptions about self-consistency or the lack thereof. We provide two concrete algorithms solving our problem, one assuming a self-consistent unit length and the other assuming a self-consistent compass. (A self-consistent compass means that each robot has its own, unchanging notion of north.) We expect the techniques developed for these cases to also have applications elsewhere in the field.

2 Preliminaries

In this section, we define our robot model and the task of Distinct Gathering under RoundRobin. We end with the nonstandard assumption of nonrigid movement.

Definition 1 (Gathering and Distinct Gathering)**.** Gathering *is the task of gathering n robots, scattered arbitrarily across the Euclidean plane. A gathering is achieved if all robots stay co-located in a single point – which one is irrelevant.* Distinct Gathering *is the same task with the additional guarantee that all robots occupy different locations in the initial configuration. We call a configuration with this property that no two robots share a location a* distinct *configuration.*

An algorithm is said to solve *a task under a given scheduler (see Definition 3 below) if it solves it for every contained schedule.*

Definition 2 (Robot Model)**.** *The robots are all identical (not even distinguished by any internal ID) and run the same algorithm. They can see and move only when activated by an external scheduler. Upon activation a robot executes a so-called Look-Compute-Move cycle:*

Look-Compute-Move Cycle. The robot first detects the current locations of all other robots. Then it uses this information exclusively to decide without further restrictions where it would like to move and what trajectory to follow. (Note that only the chosen target is relevant under rigid movement. The trajectory becomes important for nonrigid movement. However, we even restrict ourselves to robots moving on straight lines toward the computed target for our upper bounds, which yields stronger results.) Finally, the robot moves to its target along the pre-computed path, ignoring everything else. Under the standard assumption of rigid *movement, the robot always reaches its target, which concludes the Look-Compute-Move cycle.*

Inconsequential Collisions. The movement path may run across other robots without consequence; the robots neither crash nor notice such transitory encounters.

Transparent Robots. The robots are transparent in the sense that a robot is not blocked from detecting another robot by a third robot robot between them.

Inconsistent Maps. Our robots are located in the Euclidean plane and assumed to be dimensionless; they can thus be represented by points. The robots perceive the locations of the other robots precisely, but only relative to a Cartesian map *with the observing robot at the*

origin. Importantly, there is no guarantee for a consistent unit measurement, compass, or orientation among the robots. (Gathering would be trivial with fully consistent maps.) Moreover, we do not assume, by default, any self-consistency either. In other words, an adversary may arbitrarily scale, rotate, and mirror the axes of the Cartesian map perceived by the robot upon each activation without the robot noticing it. (Having such volatile maps is sometimes referred to as *variable disorientation*, as opposed to *fixed disorientation*, where each robot is assigned its own arbitrary map that then remains persistent throughout the algorithm's execution. Unfortunately, many publications considering disoriented robots remain ambiguous whether the disorientation is fixed or variable and whether it extends to the compass or the unit distance or both. A main contribution of this paper is that these distinctions are crucial.)

No Multiplicity Detection. Moreover, we assume that the robots cannot detect multiplicity, that is, whether any perceived location is shared by multiple robots or not. This includes the location of the observing robot itself; it cannot sense directly whether or not it is accompanied.

No Memory. We emphasize that the robots are stateless *(often referred to as* oblivious*): They possess no persistent memory and can thus only use the location information gathered upon activation for computing their movement path, and as soon as a robot stops moving, it forgets this information, too.*

We also need to describe the scheduler under which the robots operate. Despite our focus on the Round-Robin scheduler, we briefly describe two other important schedulers: The *semi-synchronous* scheduler (Ssync) is allowed to activate any subset of robots at any time synchronously. After an activation, it waits until all activated robots have stopped moving before activating the next arbitrary (perhaps identical) subset of robots. Any possible activation sequence is called a schedule; a scheduler can thus be seen as a family of possible schedules. The *fully synchronous* scheduler (Fsync) is the special semi-synchronous scheduler that always activates all robots. Schedulers are generally assumed to be *fair*, which means that every robot is activated infinitely often for any schedule contained in the scheduler. An unfair scheduler could easily prevent a gathering by just not activating some robots forever. The scheduler defined below is inherently fair.

Definition 3 (Round-Robin Scheduler)**.** RoundRobin *activates the robots one by one, according to a fixed but previously unknown order, in an ever-continuing cycle covering all robots. (For n robots,* RoundRobin *can therefore be seen as containing $n!$ possible schedules. However, the same can also be achieved with just a single schedule and appropriately chosen initial configurations of the anonymous robots.) We call any consecutive activation of all n robots a* round*. Just as the fully synchronous scheduler,* RoundRobin *is again a natural special case of the* semi-synchronous *scheduler; that is, it also waits with the activation of a robot if a previously activated robot is still moving. An implementation of a scheduler that has the power to guarantee a semi-synchronous activation of robots is likely to also have the option to guarantee a round-robin schedule.*

Finally, we distinguish between the standard model of *rigid movement* and the weaker assumption of the less reliable *nonrigid movement*, which models various adverse situations such as robots overheating or running out of energy during their movement.

Definition 4 (Rigidity Assumptions)**.** *The default assumption for mobile robots executing Look-Compute-Move cycles is that they always reach their target. This is called* rigid movement *as opposed to* nonrigid movement*, where a robot's movement may be stopped at any time by an adversary, as long as the robot has moved some minimum distance during its current cycle. This prematurely ends the robot's Look-Compute-Move cycle. (Note that without a guaranteed minimum movement distance or further assumptions, it would be trivial for an adversary to prevent the robots from gathering.) For our upper bounds, we assume the most challenging variant of nonrigid movement, where each robot has its own minimum movement distance. We denote the minimum of the minimum movement distances across all robots by δ. For lower bounds, we assume that all robots have the same minimum movement distance δ. Whenever the movement is nonrigid, we distinguish between a minimum movement distance that is* unknown *to the robots and the model where the robots all know δ.*

In the following, the default assumptions are variable disorientation (i.e., no self-consistency guarantee), no multiplicity detection, and nonrigid movement, unless otherwise stated.

3 Related Work

A quarter century ago, Suzuki and Yamashita [27] introduced in their seminal paper the robot model that served as the basis for much of the research on mobile robots during the following years. The key distinction from our work is their assumption that robots can detect how many robots occupy any location. They show how gathering can be achieved under this assumption [27, Thm. 3.4]. The main idea is to use the geometry of the initial configuration to create an arbitrary point of multiplicity, and then let all robots move to the point of highest multiplicity. However, sensor limitations may prevent multiplicity detection. Moreover, multiplicity makes it rather simple to achieve a gathering, such that most problems in this setting were resolved quickly. It has thus become the standard assumption in the subsequent papers that robots cannot detect multiplicity. We also use this as our default assumption from now on.

As Prencipe [25, Thm. 3] has shown, gathering is not feasible anymore under a semi-synchronous scheduler without multiplicity detection. Défago et al. later strengthened this lower bound from the semi-synchronous scheduler to RoundRobin [9, Thm. 14], showing that gathering $n \geq 3$ robots remains impossible even for this more predictable scheduler. (Note that gathering $n = 2$ robots is trivial under RoundRobin: let an activated robot move toward the opposite location [9, Lem. 4].)

Theorem 1 (Cf. [9, Thm. 14]). *Gathering under* RoundRobin *is infeasible for $n \geq 3$ robots under the default assumptions (Section 2), even if they also share a consistent unit length.*

Proof sketch. We briefly re-prove Theorem 1 as a preparation for later proofs. In fact, the proof presented here is stronger than the cited one [9, Thm. 14] because the proof as formulated here works even if the robots have a shared unit distance.

Assume toward contradiction that there is an algorithm $\mathcal{A}$ solving Gathering under RoundRobin. Fix an arbitrary schedule $\mathcal{R} \in$ RoundRobin and an arbitrary initial two-point configuration $\mathcal{I}$ with at least three robots.

Since $\mathcal{A}$ eventually achieves a gathering and only one robot can move per step, some two-point configuration $\mathcal{I}'$ is eventually reached where one robot, call it r_1, is unaccompanied and then moves directly to the location where all other robots are. We take $\mathcal{I}'$ as our initial configuration and assume that all robots perceive the same unit distance as the unaccompanied robot perceived it in the configuration $\mathcal{I}$ considered before. Since there is also no multiplicity detection, the robots thus perceive the same situation for the right adversarial choice of the orientation upon each activation of each robot. Specifically, we let all robots in the same location have the same map, and the maps of opposite locations are the same except for a rotation by π. This means that any robot observing a two-point configuration $\mathcal{I}'$ must move to the opposite location because that is what the unaccompanied robot in the previously considered two-point configuration $\mathcal{I}$ did. Now, we let RoundRobin assign a different schedule $\mathcal{R}'$ that first activates, instead of the robot r_1, which is unaccompanied in $\mathcal{I}'$, an arbitrary one of the accompanied robots, say r_2, and r_1 immediately after this. The remaining robots may be activated in any order. We can fix the remaining order to $r_3, \dots, r_n$, for example. Note that after the first move of r_2 to the previously unaccompanied r_1 – say, from position p to position q – r_1 is accompanied. Then r_1 moves to p, leaving r_2 unaccompanied. Then all remaining robots move from p to q, eventually leaving r_1 on p unaccompanied but r_2 on q accompanied. This entire process is then iterated, with p and q flipped after every iteration. This means that only accompanied robots are ever activated, which prevents a gathering. □

Note that the proof of Theorem 1 relies on the assumption of an initial configuration with accompanied robots. The natural follow-up question was therefore to ask whether this impossibility still holds for *Distinct* Gathering under RoundRobin, which guarantees that the robots all occupy pairwise distinct locations at the beginning. The case $n = 2$ is again trivial because any algorithm for Gathering works for Distinct Gathering as well. Distinct Gathering under RoundRobin becomes far more interesting and challenging for $n \geq 3$, however. Whether it is feasible or not has in fact remained a prominent open question in the field for over two decades by now, raised first at the International Symposium on Distributed Computing [8]. It has been tackled repeatedly with little success. Most recently, a paper by Terai et al. made some progress by showing how to solve the problem at least using the quite strong additional assumptions that the robots have some internal memory, share a unit distance (i.e., all perceive the same consistent unit length), and that all movements are rigid [28, Sect. 4, Alg. 10]. However, the general consensus has remained that Distinct Gathering is most likely impossible under RoundRobin under the standard assumptions outlined above. A conjecture by Défago et al. makes this explicit [9, Conj. 2].

Conjecture 2 (Cf. Défago et al. [9, Conj. 2])**.** *There is no algorithm solving Distinct Gathering under* RoundRobin *for $n \geq 3$ robots under the default assumptions detailed in Section 2.*

They corroborate their assumption by proving [9, Thm. 16] the impossibility of Distinct Gathering under RoundRobin under the assumption that any algorithm solving this problem is a so-called *rapid-gathering* algorithm [9, Conj. 1].

Definition 5 (Rapid-gathering algorithm, cf. Défago et al. [9, Def. 1])**.** *Any algorithm is a* rapid-gathering *algorithm if there is an initial configuration of n robots in distinct locations and a Round-Robin activation order such that the algorithm gathers all robots within $n - 1$ activations.*

Conjecture 3 (Cf. Défago et al. [9, Conj. 1])**.** *Every algorithm that solves Gathering under* RoundRobin *is a rapid-gathering algorithm.*

They further substantiate this plausible claim by considering a series of convincing examples.

We finally mention that Flocchini et al. [11, 12, Thms. 2,3] recently considered centralized schedulers, a superclass that, unlike RoundRobin, does not guarantee a fixed activation order. They show the possibility of Self-Stabilizing Gathering under these schedulers with the additional assumption of *weak multiplicity* and the impossibility without it. Note, however, that even more general versions of both statements had been known already [16, Thms. 9, 10, 13, 14].

4 First Contribution: Resolving the Long-Standing Open Conjecture

In this section we fully resolve the open question about the feasibility of Distinct Gathering under RoundRobin, giving an unexpected answer, which might seem contradictory at first sight.

We first prove in Section 4.1 that the problem is indeed impossible to solve for the case of $n = 3$ but then show in Section 4.2 that it becomes feasible again for four and more robots, even though these robots have to pass through three-point configurations during the execution of their algorithm. We address this apparent paradox after presenting the proofs.

4.1 Distinct Gathering Is Impossible for Exactly Three Robots

We show that gathering three robots is infeasible under RoundRobin even if the initial configuration is guaranteed to be distinct. We remark that this result unifies and strengthens two separate impossibility results by Défago et al. [9, Thms. 14 and 15] at the same time for $n = 3$, extending one from Gathering to *Distinct* Gathering and the other from so-called k-bounded schedulers to the far more predictable RoundRobin scheduler.[1]

Theorem 4. *Distinct Gathering under* RoundRobin *is impossible for $n = 3$ robots under the default assumptions detailed in Section 2.*

The full, rigorous proof for the theorem is provided in the full version of this paper [17].

[1] We note that the cited paper implicitly assumes neither consistency nor self-consistency in the maps of the robots, not even for the unit distance, as evidenced by the remark in the proof of Theorem 14 that all two-point configurations are indistinguishable. For the cited proofs, this assumption is insubstantial and easily removed as we demonstrate with Theorem 1. In other cases, however, it is crucial whether the perceived unit distance is self-consistent for each robot, as we show with the contrast provided by the lower bound of Theorem 4 and the upper bounds of Theorems 6 and 7.

Proof sketch. Fix an algorithm $\mathcal{A}$, a schedule $\mathcal{R}$, and an initial configuration $\mathcal{I}$.

A simple but important proof ingredient is that for every two-point configuration we can assume a *normalized view*: Any robot perceives the other location to be at distance 1 exactly north, letting all robots behave the same way from their local perspectives.

We consider $\mathcal{A}$'s last transition from some three-point configuration $\mathcal{I}'$ to a two-point configuration. We use the normalized view and the fact that the algorithm does not revert to a three-point configuration multiple times to derive a contradiction. At some intermediate point, we need to consider $\mathcal{I}'$ as an alternative initial configuration with an alternative schedule $\mathcal{R}'$, on which the algorithm must also achieve a gathering, to obtain further properties about $\mathcal{I}$ and $\mathcal{R}$. These additional properties allow us to eventually obtain the required contradiction. □

We have proved Distinct Gathering under RoundRobin to be impossible for $n = 3$, confirming Conjectures 2 and 3 in this case. The natural assumption is that this result now extends beyond three robots to the case of $n \geq 4$ because gathering more robots seems to only increase the difficulty of the problem. Indeed, any initial distinct configuration with four or more robots is forced to pass through a three-point configuration under RoundRobin before a gathering can be achieved. Surprisingly, we can show that this argumentation is flawed by providing an algorithm that solves the problem for any given $n \geq 4$. This disproves the established Conjectures 2 and 3 and resolves the long-standing open question about the feasibility of Distinct Gathering under RoundRobin. A detailed discussion of why the seeming paradox between our discovery and the line of reasoning outlined above does not yield a contradiction is found in the full version of this paper [17].

4.2 Distinct Gathering Is Possible for Four and More Robots

We provide in this section an algorithm that solves Distinct Gathering under RoundRobin for four or more robots. Moreover, we design this algorithm in a robust way such that it works not only under the already minimalistic assumptions underlying the cited conjecture, which coincide with the default model presented in this paper, but also if the robots are allowed to move only in straight trajectories and even if they suffer from nonrigid movement, which may unpredictably stop them in their tracks at any time.

Theorem 5. *Distinct Gathering under* RoundRobin *can be solved for any given $n \geq 4$. This result holds even for robots restricted to moving to their target in a straight line and under the assumption of nonrigid movement with an unknown minimal movement distance.*

The full proof, including the full pseudo-code and the proof of correctness, is deferred to the full version of this paper [17]. We provide here only a brief overview of the algorithm and some key proof ideas.

The algorithm proceeds through a sequence of nine phases (potentially with repetitions), relying on geometric properties—especially isosceles right triangles—to ensure convergence and the eventual gathering. We refer to Figure 1 for a visual representation of each phase and the possible transitions between them. We now describe the individual phases in some more detail.

Phase 1 ensures that the configuration contains at most one isosceles right triangle or multiple such triangles sharing a common vertex. This is done by moving robots on isosceles right triangles to destroy them, while avoiding so-called forbidden points, which would create new triangles or result in collisions. Phase 2 ensures the existence of *exactly one* isosceles right triangle or multiple such triangles sharing a common vertex by letting the robots closest to forming such a triangle move, while others wait. Phase 3 then ensures that there is just a single isosceles right triangle by having the common robot remain static, while others move slightly to break the surplus isosceles right triangles one by one. The first three phases all maintain distinctness.

In Phase 4, the robots not on the unique right isosceles triangle move to the endpoints of its base by targeting the closest endpoint. This keeps the triangle invariant and the robot at the right-angle vertex remains unaccompanied. In Phase 5, the robot at the right-angle vertex simply moves to the midpoint of the base. In Phase 6, the remaining robots start moving to the midpoint, where a single robot has just arrived at the end of the previous phase. Phase 6 ends as soon as the configuration is no longer a midpoint triple, i.e., a three-point configuration with one point exactly in the middle between the other two. The phase can end in a four-point configuration, which leads to Phase 7. Another possibility is a two-point configuration, which occurs if the last robot of either endpoint moves directly to the midpoint and halves the diameter, leading to Phase 9. A crucial property in this case is that all robots at the former midpoint have just moved in sequence and will thus be activated again only after all the robots at the other endpoint have been activated. The last possibility is a non-midpoint triple three-point configuration, which occurs if the last robot of an endpoint is prematurely stopped after moving at least δ, which reduces the diameter by at least δ and leads to Phase 8.

The purpose of Phase 7 is to return to Phase 4 after a failure caused by non-rigid movement. This is done by the non-aligned robot moving to create a new right isosceles triangle of a diameter reduced by at least $\delta/2$ compared to any previous iteration of Phase 4, which prevents an infinite cycle. Phase 8 corrects a configuration with three collinear but imbalanced robots by letting the middle one move to the correct midpoint. This means a return to Phase 6 with a diameter reduced by at least δ. The action in Phase 9 is simple: In a two-point configuration, any activated robot targets the opposite point. Thanks to the crucial property described above for the transition from Phase 6 this either results in a gathering or, due to nonrigid movement, in a new three-point configuration. The latter means a return to Phase 6 or 8, with a guaranteed diameter reduction of at least δ.

We can verify that each phase terminates in a finite number of steps. Moreover, the only cyclical returns occur due to nonrigid interruptions. Since each cycle iteration reduces the configuration diameter by at least δ, all moves are eventually rigid, ensuring that a gathering occurs in Phase 9.

Another key ingredient in the proof of phase termination besides the parameter reduction is the following observation: Whenever in Phase 4, the robot at the right angle is unaccompanied, then it moves down in Phase 5 and is the last one that has moved when Phase 6 starts. This is the main property that makes it safe to enter a three-point configuration and allows, assuming rigid movement,

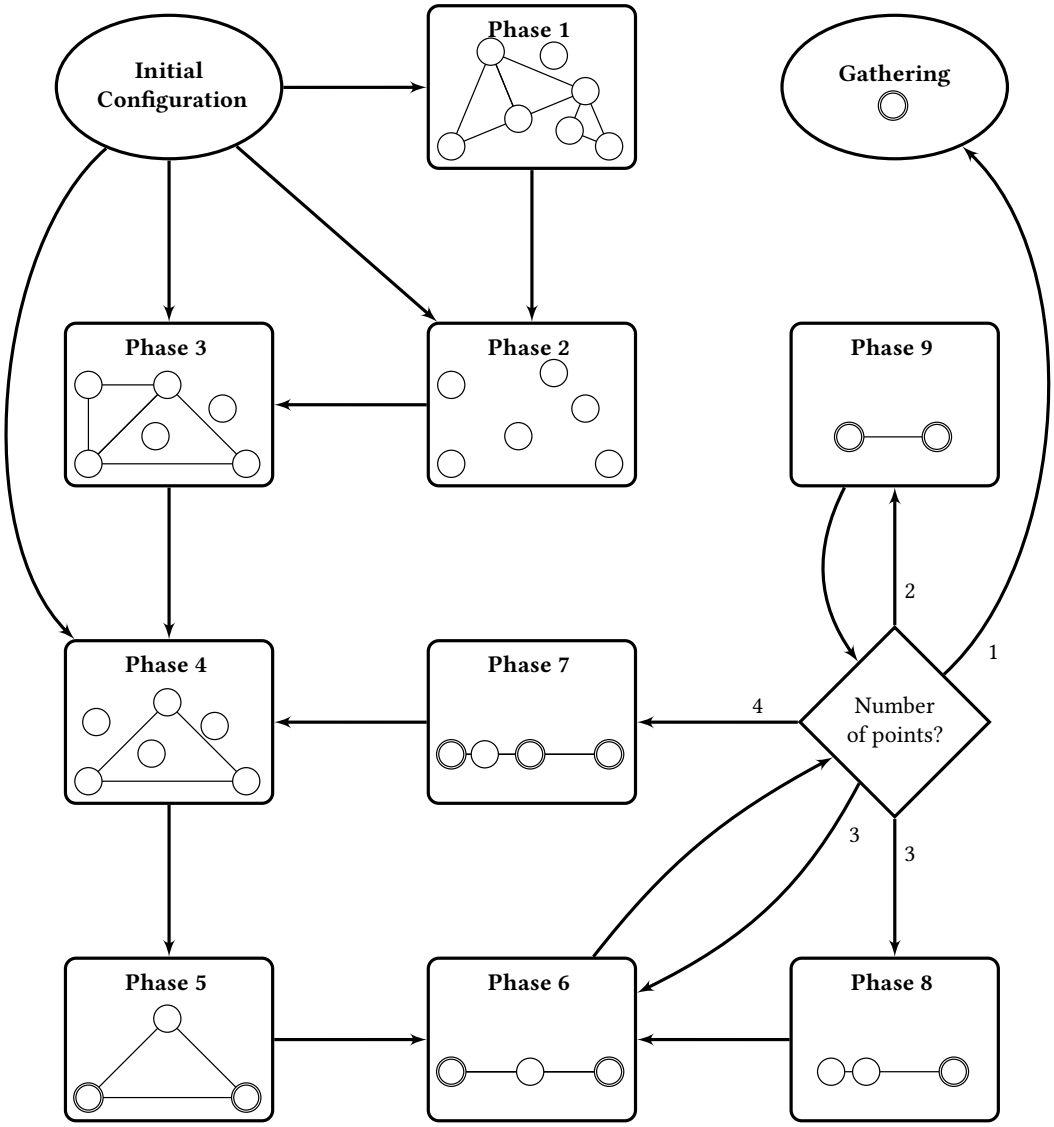


Figure 1: Phase transition diagram. The illustrations show typical configurations that can occur at the start of each phase. The arrows show which phases may lead to which. It is proven that no infinite loops occur.

for a gathering as soon as all other robots have been activated once more.

On a very high level, the crucial difference to the case of $n = 3$ robots and the impossibility proof of Theorem 4 is thus that the additional robots can be used to store, in the geometric configuration itself, information about the adversarially chosen activation order of the robots, which then lets the algorithm enter the dangerous three-point configurations in a controlled manner, which is impossible with only three robots. A more detailed discussion of this point is found in in the full version of this paper [17].

We now turn to the second main message of this paper, proving that self-consistency can be crucial for successful coordination.

5 Second Contribution: Proving the Value of Self-Consistency

We first consider the model with only a self-consistent unit length in the following section and then the model with only a self-consistent compass.

5.1 Assuming Only a Self-Consistent Unit Length

As already mentioned in the introduction, the last attempt by Terai et al. [28, Sect. 4, Alg. 10] to attack the open problem resolved in this paper tried to approach the problem from above by trying to find a minimal set of assumptions under which Distinct Gathering becomes feasible. They managed to find an algorithm for

the problem if they grant the algorithm some internal memory (referred to as internal lights) and at the same time a shared unit distance. With our main result, Theorem 5, we have already shown that indeed no additional assumptions at all are needed for four or more robots, just as for the trivial cases of one and two robots. For three robots, however, we proved with Theorem 4 that the problem unexpectedly cannot be solved in the intermediate case of three robots without additional assumptions. To provide a complete picture of Distinct Gathering under RoundRobin, we would like to reduce the additional assumptions down to the minimum. Indeed, we can show that the aforementioned result still made several assumptions that can be removed. Specifically, we prove that solving the problem requires neither internal nor external memory, nor even the assumption of a consistent unit length. We present an algorithm that works with only the additional assumption that the robots have a self-consistent length measurement. Each robot may be adversarially assigned its own unit distance for any given initial configuration. We now state this result formally.

Theorem 6 (Possibility of Distinct Gathering for $n = 3$ and Inconsistent but Self-Consistent Unit Measurement). *Distinct Gathering under* RoundRobin *is feasible for* 3 *robots with full disorientation except for a possibly inconsistent but self-consistent distance unit for each robot and under the otherwise default assumptions detailed in Section 2.*

The full proof is found in the full version of this paper [17]. We provide here an outline of the main ideas.

Proof Sketch. The algorithm, also found in the full version of this paper [17], relies on a distance classification scheme. Specifically, the continuum of positive real numbers $\mathbb{R}_{>0}$ is partitioned into two disjoint infinite unions of intervals: $\mathbb{D}_{\text{golden}} = \bigcup_{k\in 2\mathbb{Z}}[\varphi^{k+1}, \varphi^k)$ and $\mathbb{D}_{\text{gather}} = \bigcup_{k+1\in 2\mathbb{Z}}[\varphi^{k+1}, \varphi^k)$, where $\varphi = \frac{2}{1+\sqrt{5}}$ is the golden ratio. These intervals are logarithmically spaced and serve to discretize distance observations in a self-consistent manner.

The primary challenge is the lack of consistency in perceived distances: each robot uses its own arbitrary unit length. Consequently, direct reliance on any single robot's perception to define globally consistent behavior is impossible. The key idea is to enforce a structure on the observed distances that causes predictable transitions between configurations.

The algorithm moves through the following high-level phases:

(1) **Initialization and Contraction:** Starting from any arbitrary configuration (collinear or non-collinear), the robots execute moves that eventually reduce the diameter of the configuration without ever increasing it. Through repeated activations and based on the classification of the observed triangle's side lengths, the algorithm gradually transforms any initial triangle into one with a unique shortest side.
(2) **Controlled Symmetry Breaking:** Once a triangle with a unique shortest side is achieved, the robot not adjacent to this side is eventually activated and may attempt to move toward the other robots. However, this action is taken only when the perceived length of the shortest side falls into $\mathbb{D}_{\text{golden}}$. This restriction ensures that the two-point configuration created is well-structured and predictable in terms of how the observing robot will behave.
(3) **Two-Point to Three-Point Cycling:** When a two-point configuration is created, the robot that initiated the move is now collocated with one of the other robots, while the third robot is at a different location. The key insight is that based on the perceived distance (again classified into $\mathbb{D}_{\text{golden}}$ or $\mathbb{D}_{\text{gather}}$), robots either remain stationary or move in small, controlled steps to create specific collinear three-point configurations. This cycling between two-point and three-point formations allows the robots to coordinate indirectly through geometry and ensures that enough variations of distance configurations are explored.
(4) **Eventual Rigid Movement:** Another critical component of the algorithm is that it ensures a bounded decrease of the configuration diameter after a finite number of steps. Because the system allows only movements that do not increase the diameter, and the distance thresholds used in the classification scheme are strictly decreasing with powers of φ, the configuration's diameter eventually becomes smaller than any possible non-rigid movement threshold. This guarantees that we can thereafter assume all robot moves to be rigid.
(5) **Convergence to Gathering:** After sufficiently many transitions between specific two-point and three-point configurations, the robots arrive at a configuration where all three perceive distances from $\mathbb{D}_{\text{gather}}$ and are activated in one of a bounded number of scenarios. Each of these is explicitly handled in the algorithm and shown to lead to gathering within a finite number of steps, regardless of the scheduler's choices or initial scale mismatches. □

5.2 Assuming Only a Self-Consistent Compass

We have found with the self-consistent unit length a clearly minimal assumption to solve the problem. It is counterintuitive that a purely self-consistent property can help the robots coordinate among each other at all. The conventional assumption is that some shared knowledge or perception is required for the robots to improve their capabilities. Motivated by the previous result, we have also investigated the case where not the unit length, but instead a sense of direction, called compass (essentially a vector of variable length always pointing in the same direction), is given to the robots in a self-consistent way. As it turns out, the problem is even solvable under this assumption, without any self-consistency for the unit length, as stated in following theorem. The techniques for the two proofs are quite dissimilar. It seems impossible for the algorithms to simulate each other.

Theorem 7 (Possibility of Distinct Gathering for $n = 3$ and an Inconsistent but Self-Consistent Compass). *Distinct Gathering under* RoundRobin *is possible for* 3 *robots with full disorientation except for a possibly inconsistent but self-consistent compass for each robot and under the otherwise default assumptions detailed in Section 2.*

The full proof is found in the full version of this paper [17]. We provide here an outline of the main ideas.

Proof Sketch. The algorithm, also deferred to the full version of this paper [17], solves the Distinct Gathering problem under

RoundRobin for three robots with self-consistent compasses, even when faced with nonrigid movement.

The overall strategy of the algorithm is structured into two main phases. The first phase achieves a collinear configuration in which all robots possess a well-defined and consistent sense of direction along the shared line. A robot derives such a sense of direction from its self-consistent compass vector if this vector is not orthogonal to the line formed by the robot positions. It is proven that such a collinear configuration with a self-consistent sense of direction for all robots eventually emerges from any initial configuration. Furthermore, we can show that the diameter of the configuration decreases over time as long as nonrigid movements occur, thus guaranteeing that rigid movement can eventually be assumed without loss of generality.

Once a collinear configuration as described is reached, the second phase of the algorithm begins. This phase relies on the robots' ability to break symmetry using their self-consistent compasses. The robots restrict their movements to the line formed by them and behave differently depending on their observed relative positions and compass directions. The correctness of this phase is verified through an exhaustive case analysis of all 48 possible configurations demonstrating that they eventually lead to a gathering.

Together, these two phases ensure that the algorithm terminates with all robots gathered at a single location. □

6 Conclusion

By showing that Distinct Gathering is possible under RoundRobin in our default model for all robot numbers except for three robots, we have resolved a long-standing question about mobile robots, disproving the earlier conjectures.

Moreover, we have explored the minimal assumptions required to make the case $n = 3$ feasible. The last algorithm that attempted to minimize the assumptions still required rigidity of movement, a common unit distance, and, additionally, memory. We designed an algorithm that reduces all of this to only a self-consistent unit length, and a second algorithm that only needs a self-consistent compass. This highlights the crucial importance of this often overlooked distinction between fully volatile and self-consistent perceptions. Our results thus also prove in our context that self-consistency is a virtue enabling collaboration even in the absence of any other shared values.

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